



11245 - Late-time spectral energy distribution of the first off-nuclear tidal disruption event AT 2024tvd

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Kishore C. Patra (PI)	University of California - Santa Cruz
Nicholas Earl (CoI)	University of Illinois at Urbana - Champaign
Prof. Ryan Foley (CoI) (US Admin CoI)	University of California - Santa Cruz
Dr. Samaporn Tinyanont (CoI)	National Astronomical Research Institute of Thailand
Kyle Davis (CoI)	University of California - Santa Cruz
Dr. Enrico Ramirez-Ruiz (CoI)	University of California - Santa Cruz
Prof. Ashley Villar (CoI)	Harvard University
Dr. Sebastian Gomez (CoI)	University of Texas at Austin
Dr. K. Decker French (CoI)	University of Illinois at Urbana - Champaign
Dr. Kirsty Taggart (CoI)	Universities Space Research Association
Prasiddha Arunachalam (CoI)	University of California - Santa Cruz
Dr. Phillip Macias (CoI)	University of California - Santa Cruz
Ravjit Kaur (CoI)	University of California - Santa Cruz

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSpec IFU spectroscopy				
	1	NIRSPEC IFU 2024tvd	NIRSpec IFU Spectroscopy	(1) 2024tvd
NIRCam Imaging				
	2	NIRCam 2024tvd	NIRCam Imaging	(1) 2024tvd

ABSTRACT

Tidal disruption events (TDEs) are luminous flares produced when a star is torn apart and accreted by a massive black hole (MBH). More than 100 TDEs have been discovered so far, almost all originating from galactic nuclei where MBHs typically reside. However, galaxy-merger simulations predict a substantial population of wandering MBHs outside galactic centers--yet no TDE had ever been associated with one until now. The discovery of AT 2024tvd, the first confirmed off-nuclear TDE, opens an entirely new window on this phenomenon. Its location, offset from the bright stellar background of the galactic nucleus, offers a uniquely clean view of accretion physics in action and a chance to test the long-standing idea that TDEs arise from stars bound in nuclear star clusters (NSCs).

We propose coordinated multiwavelength observations with JWST (infrared; NIRSpec IFU + NIRCам), HST (UV + optical imaging), and XMM-Newton (X-ray) to obtain a late-time (>700 days after optical peak) spectral energy distribution (SED) spanning X-ray to infrared wavelengths. The SED will be modeled with three components--the accretion disk, an NSC, and a dust echo--to constrain the MBH mass, accretion rate, and disk size. The NSC component will directly test the presence of a bound stellar cluster and measure its mass, age, and metallicity, testing whether the wandering MBH was delivered through a past galaxy merger. The dust component will characterize the temperature and luminosity of the surrounding dust. These observations will provide unique insights into the progression of accretion in TDEs, in addition to the demographics and growth pathways of wandering MBHs.

OBSERVING DESCRIPTION

NIRSpec IFU spectroscopy will be performed using three disperser-filter combinations to cover the full NIRSpec wavelength range:

G140H + F100LP

G235H + F170LP

G395H + F290LP

NIRCам imaging will be conducted simultaneously in both the SW and LW channels using the following filters:

F150W

F200W

F277W

F444W

Proposal 11245 - Targets - Late-time spectral energy distribution of the first off-nuclear tidal disruption event AT 2024tvd

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	2024tvd	RA: 17 10 42.5700 (257.6773750d)	Epoch of Position: 2000	
			Dec: +28 50 14.99 (28.83750d)		
			Equinox: J2000		
	<i>Comments:</i> <i>Category=Unidentified</i> <i>Description=[Infrared sources, Ultraviolet sources, Variable radiation sources, Visible sources, X-ray sources]</i> <i>Extended=NO</i>				

Proposal 11245 - Observation 1 - Late-time spectral energy distribution of the first off-nuclear tidal disruption event AT 2024tvd

Observation	Proposal 11245, Observation 1: NIRSPEC IFU 2024tvd											Thu May 28 15:00:32 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	2024tvd	RA: 17 10 42.5700 (257.6773750d) Dec: +28 50 14.99 (28.83750d) Equinox: J2000				Epoch of Position: 2000					
	Comments: Category=Unidentified Description=[Infrared sources, Ultraviolet sources, Variable radiation sources, Visible sources, X-ray sources] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2	15	3	false	true	NONE	4	12	13305.068	273394
	2	G235H/F170LP	NRSIRS2	15	3	false	true	NONE	4	12	13305.068	273394
	3	G395H/F290LP	NRSIRS2	10	1	false	true	NONE	4	4	2976.134	273394

Proposal 11245 - Observation 2 - Late-time spectral energy distribution of the first off-nuclear tidal disruption event AT 2024tvd

Observation	Proposal 11245, Observation 2: NIRCam 2024tvd										Thu May 28 15:00:32 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
	Comments: Target centered onto B1										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(1)	2024tvd	RA: 17 10 42.5700 (257.6773750d) Dec: +28 50 14.99 (28.83750d) Equinox: J2000				Epoch of Position: 2000				
	Comments: Category=Unidentified Description=[Infrared sources, Ultraviolet sources, Variable radiation sources, Visible sources, X-ray sources] Extended=NO										
Template	Module						Subarray				
	B						FULL				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		SMALL-GRID-DITHER				3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F277W	BRIGHT2	4	1	12	12	1030.73	273394	
	2	F200W	F444W	BRIGHT2	4	1	12	12	1030.73	273394	
Special Requirements	Offset 35.0 arcsec, 35.0 arcsec										